# Aniket Rege

Department of Computer Sciences
University of Wisconsin-Madison
WI 53703

aniketr@cs.wisc.edu
Google Scholar
Personal Website

## Education

**University of Wisconsin-Madison** *2023 - present*
*PhD in Computer Sciences*
Advisor: Prof. Ramya Korlakai Vinayak & Prof. Yong Jae Lee

**University of Washington** *2021 - 2023*
*MS in Electrical and Computer Engineering*
Advisor: Prof. Ali Farhadi

**National Institute of Technology Karnataka** *2014 - 2018*
*B.Tech in Electronics and Communication Engineering*
Advisor: Prof. Muralidhar Kulkarni

## Work Experience

**Google DeepMind, Bangalore** *June - Nov, 2026*
*Student Researcher, GenAI*
Advisors: Gagan Madan & Simon Baumgartner

**Meta Reality Labs, Redmond** *May - Dec, 2025*
*Research Scientist Intern, Reality Labs*
Advisors: Dr. Hyo Jin Kim & Dr. Yuning Chai

**NVIDIA, Redmond** *June - Sept, 2022*
*Applied Research Intern, Deep Learning for Autonomous Vehicles*
Advisors: Dr. Nikolai Smolyanskiy & Dr. Stan Birchfield

**Samsung Research India, Bangalore** *2018 - 2021*
*AI Camera Research Engineer, Visual Intelligence Group*
Advisors: Sudha Velusamy & Dr. Shankar Venkatesan

## Publications

### Preprints

* equal contribution, † equal advising

1. **Playing with Fire: What Transfers When RL Trains a Language Agent?**
Mahesh Ramesh, Kaousheik Jayakumar, Hemanth Ram, Pavan Thodima, Ramani Duraiswami, Dinesh Manocha, **Aniket Rege**†, Manolis Vlatakis†
*Reinforcement Learning from World Feedback Workshop @ ICML, 2026.*
*Efficient Reasoning Workshop @ COLM, 2026* (**Spotlight**).
*Under review @ ICLR, 2026.*

### Conference Publications

9. **Sparks of Cooperative Reasoning: LLMs as Strategic Hanabi Agents.**
Mahesh Ramesh, Aswinkumar Ramkumar, Pavan Thodima, Kaousheik Jayakumar, **Aniket Rege**†, and Manolis Vlatakis†.
*International Conference on Machine Learning (**ICML**), 2026.*
- *Knowledge-Intensive Multimodal Reasoning Workshop @ ICCV, 2025.*
- *Bridging Language, Agent, and World Models Workshop @ NeurIPS, 2025.*
- *Scaling Environments for Agents Workshop @ NeurIPS, 2025.*
- *Multi-Turn Interactions in Large Language Models Workshop @ NeurIPS, 2025.*

8. **Agentic Very Long Video Understanding.**
**Aniket Rege**, Arka Sadhu, Yuliang Li, Kejie Li, Ramya Korlakai Vinayak, Yuning Chai, Yong Jae Lee, and Hyo Jin Kim.
*The 64th Annual Meeting of the Association for Computational Linguistics (**ACL**), 2026* **(Oral)**.

7. **CuRe: Cultural Gaps in the Long Tail of Text-to-Image Models**.
**Aniket Rege**, Zinnia Nie, Mahesh Ramesh, Unmesh Raskar, Zhuoran Yu, Aditya Kusupati†, Yong Jae Lee†, and Ramya Korlakai Vinayak†.
*IEEE/CVF International Conference on Computer Vision (**ICCV**), 2025.*
- *Demographic Diversity in Computer Vision workshop @ CVPR, 2025* (**Oral**).
- *Critical Evaluation of Generative Models and Their Impact on Society Workshop @ ICCV, 2025* (**Oral**).
- *What is Next in Multimodal Foundation Models? Workshop @ ICCV, 2025.*
- *Safe and Trustworthy Multimodal AI Systems Workshop @ ICCV, 2025.*

6. **PAL: Sample-Efficient Personalized Reward Modeling for Pluralistic Alignment**.
Daiwei Chen, Yi Chen, **Aniket Rege**, Zhi Wang, and Ramya Korlakai Vinayak.
*International Conference on Learning Representations (**ICLR**), 2025.*
- *Stereotypes Across Cultures in Language Technologies workshop @ ACL, 2026* (**Oral**).
- *Models of Human Feedback for AI Alignment workshop @ ICML, 2024* (**Oral**).
- *Workshop on Behavioral Machine Learning @ NeurIPS, 2024* (**Spotlight**).
- *Pluralistic Alignment workshop @ NeurIPS, 2024.*

5. **AdANNS: A Framework for Adaptive Semantic Search**.
**Aniket Rege***, Aditya Kusupati*, Sharan Ranjit S, Alan Fan, Qingqing Cao, Sham Kakade, Prateek Jain, and Ali Farhadi.
*Neural Information Processing Systems (**NeurIPS**), 2023.*
- *Practical ML for Developing Countries Workshop* @ ICLR, 2023 (**Oral**).

4. **Matryoshka Representation Learning**.
**Aniket Rege***, Aditya Kusupati*, Gantavya Bhatt*, Matthew Wallingford, Aditya Sinha, Vivek Ramanujan, William Howard-Snyder, Kaifeng Chen, Sham Kakade, Prateek Jain, and Ali Farhadi.
*Neural Information Processing Systems (**NeurIPS**), 2022.*
- *Vision Transformers: Theory and Applications workshop @ NeurIPS, 2022* (**Oral**).
- *Self-Supervised Learning - Theory and Practice workshop @ NeurIPS, 2022.*
- *Computer Vision in the Wild workshop @ ECCV, 2022.*

3. **Spatio-Temporal Video Representation Learning for AI Based Video Playback Style Prediction**.
Rishubh Parihar*, Gaurav Ramola*, Ranajit Saha, Raviprasad Kini, **Aniket Rege**, and Sudha Velusamy.
*Structured Representations for Video Understanding workshop @ **ICCV**, 2021.*

2. **FabSoften: Face beautification via dynamic skin smoothing, guided feathering, and texture restoration**.
Sudha Velusamy, Rishubh Parihar, Raviprasad Kini, **Aniket Rege**.
*IEEE/CVF Conference on Computer Vision and Pattern Recognition workshops (**CVPR W**), 2020.*
- *New Trends in Image Restoration workshop @ CVPR, 2020.*

1. **QUIC Protocol Performance in Wireless Networks**.
Prashant Kharat*, **Aniket Rege***, Aneesh Goel*, and Muralidhar Kulkarni.
*IEEE International Conference on Communication and Signal Processing (**ICCSP**), 2018.*

## Patents

3. **Method and electronic device for detecting candid moment in image frame**.
**Aniket Rege***, Gaurav Ramola*, Nikhar Maheshwari*, Sudha Velusamy, Girish Kulkarni, Sai Pranav Mathivanan, Swadha Jaiswal, and Pradeep Kumar Vagireddi.
*U.S. Patent 12,299,959, issued May 13, 2025.*

2. **Method and electronic device for processing facial images**.
Sudha Velusamy, Rahul Varna, Raviprasad Kini, **Aniket Rege**, Rishubh Parihar, Daeyoung Hyun.
*U.S. Patent 11,335,122, issued May 17, 2022.*

1. **Mobile Accessory For 3D Selfie Creation**.
**Aniket Rege***, Shankar Venkatesan*, Ankit Jalan*, Mynepalli Siva Chaitanya, Arko Sabui, Rishi Franklin, Viswanath Veera, Abhijeet Singh.
*IN 411,097, issued March 23, 2018.*

## Theses

2. **Congestion Control in Wireless Networks**.
**Aniket Rege***, Aneesh Goel*, and Muralidhar Kulkarni.
*Undergraduate Thesis, Electronics and Communication Engineering, NITK Surathkal, 2018.*

1. **MRL-AdANNS: Matryoshka Representation Learning for Web-Scale Adaptive Semantic Search**
Aniket Rege
*Masters Thesis, Electrical and Computer Engineering, University of Washington, 2023.*

## Selected Awards and Honors

- **CS Departmental Scholarship**, UW-Madison *2023 - present*
- **NeurIPS Scholar Award** *2022*
- **University of Washington Graduate School Award** for NeurIPS '22 *2022*
- **Samsung HQ Best Project Award** as founding member of **Single Take Camera**, the USP of the Galaxy S20 smartphone *2021*
- **Samsung Best Paper Award '19 Finalist** for FabSoften, later published at CVPR '20, selected from all worldwide offices *2019*
- **Samsung India Outstanding Contribution Award** for shipping **AI Beauty Solution** to 50M+ smartphones worldwide *2019*
- **Road to Global Entrepreneurship Summit (GES) '17 national finalist** for *An Analysis of Sleep Hygiene*. Conducted by NITI Aayog under the Government of India. *2017*

## Professional Responsibilities

- *Reviewing*: ICCV/ECCV (2023 - present), NeurIPS (2024 - present), ICLR (2024 - present), ICML (2025 - present), CVPR (2025 - present)
- *Workshop Organization*
  - Multimodal Alignment for a Pluralistic Society (MAPS) *CVPR 2026*
- *Graduate Teaching Assistant*, University of Wisconsin-Madison
  - CS 540: Introduction to Artificial Intelligence - Prof. Yudong Chen, Prof. Blerina Gkotse *Fall 2023-25*
- *Graduate Teaching Assistant*, University of Washington
  - CSE 493: Deep Learning - Prof. Ranjay Krishna *Spring 2023*
  - CSE 547: Machine Learning for Big Data - Prof. Tim Althoff *Winter 2023*
  - EE 596: Computer Vision: Classical and Deep Methods - Dr. Stan Birchfield *Fall 2021*
- *Mentorship*
  - Mahesh Ramesh
    MS CS, UW Madison *Jan '25 - May '26*
  - Unmesh Raskar
    MS ECE, UW Madison → ML Engineer @ Cadence *Jan '25 - May '25*
  - Zinnia Nie
    BS Student, UW Madison CS → MS Student @ U Michigan CS *Jun '24 - May '25*
  - Sharan Ranjit
    MS Student, UW ECE → Machine Learning Engineer @ Autodesk *Jun '22 - May '23*
  - Ankit Pandey
    Intern, Samsung R&D *Apr - Jul 2021*
  - Nikhar Maheshwari
    Intern, Samsung R&D → MS Student @ Stony Brook CS *May - Jul 2020*
- *Samsung University Representative* - Technical liason for hiring of students from NITK *2018 - 2021*
- *Undergraduate Teaching Assistantship, Linear Signals and Systems* - Department of ECE, NITK *2015 - 2016*

## Talks

- Agentic Very Long Video Understanding
  - UW Madison ML Seminar *December 2025*

  - Meta Reality Labs *August 2025*
- Cultural Gaps in the Long-Tail of Text-to-Image Models
  - The IEEE/CVF International Conference on Computer Vision (ICCV) *October 2025*
  - CEGIS workshop @ ICCV (Spotlight Talk) *October 2025*
  - Meta Reality Labs *July 2025*
  - DemoDiv Workshop @ CVPR (Spotlight Talk) *June 2025*
  - The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) *June 2025*
- Adaptive Representations for Semantic Search
  - Jina AI *March 2024*
  - Neural Information Processing Systems (NeurIPS) *December 2023*
- Matryoshka Representation Learning
  - UW Madison ECE MLOPT Idea Seminar (YouTube Link) *April 2024*
  - Jina AI *March 2024*
  - Latent Space Podcast LLM Paper Club *February 2024*
  - Kilian Q. Weinberger's Lab @ Cornell *January 2023*
  - Neural Information Processing Systems (NeurIPS) *December 2022*
- Miscellaneous
  - ISTE NITK Alumni Series - The Start of My Career (YouTube Link) *July 2020*

## Extracurriculars

**Convener and Head, Indian Society for Technical Education, NITK** *Apr 2017 - May 2018*
Advisor: Dr. Basavaraju Manu, *NITK*

- Led a team of 100+ students to conduct technical talk and cross-domain technical projects across STEM fields, business, literature, and graphic design.
- Head of the Electrical Engineering special interest group.
- Initiated and organized the university's first cross-campus hackathon during the annual tech fest with SRIB.
- Pioneered NITK's first annual technical club project expo in association with IEEE-NITK.
- Initiated the formation of the club's alumni network on LinkedIn, as well as the club's first summer mentorship program.

**Joint Convener and Head, Engineer (Technical Fest)** *Aug - Oct 2017*

- Joint head of NITK's annual technical fest, Engineer '17.
- Co-led the Technites committee, a three day expo of EECS projects primarily focused on embedded electronics.

**Performing Artist (Vocals), NITK Music Club** *Aug 2016 - May 2018*